Using the DMON2 with a Slocum glider for archival passive acoustic monitoring

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The instructions below will allow you to (A) install the software required to communicate with a DMON2 instrument, (B) load new firmware, (C) conduct a test of the firmware and hydrophone, (D) charge the DMON2, (E) prepare for deployment, (F) deploy a DMON2-equipped glider, (G) recover a DMON2-equipped glider, (H) install new _boot.bin and _reboot.bin files (only if needed), (I) decode DMON2 messages, including those needed to access duty-cycled recordings, (J) revive a dead battery, and (K) shut down a DMON2 instrument without attaching a USB cable. Section (L) provides information about the hydrophones in the DMON2 hydrophone head.

When testing, you may find it useful to have software that allows you to listen to recorded audio and view spectrograms. Raven Lite is a free program offered by Cornell University's Center for Conservation Bioacoustics that works very well for this purpose. You can find it here: https://ravensoundsoftware.com/software/raven-lite/.

Note that these instructions are intended for using the DMON2 for recording audio only. For applications using the DMON2/LFDCS system to detect sounds produced by marine mammals in near real time, collaboration with the Woods Hole Oceanographic Institution and separate instructions are required.

A. Install DMON2 communications software

1. *Do not* attach the DMON2 to the PC via the USB cable yet.
2. Unzip the dmon2 distribution in a folder on your PC.
3. Inside this folder, run the zadig_2.2.exe application.

4. Under the "Device" menu select "Create New Device". See Figure 1.
5. In the top field, enter the name of the device: “DMON_Boot”. See Figure 2.
6. In the first two fields of USB ID, enter 0451 and 9001. Leave the last USB ID field blank.
7. Above the “Install Driver” button, ensure that WinUSB is selected “WinUSB (v6.1.7600.16385)”
8. Click the “Install Driver” button.

9. Under the "Device" menu select "Create New Device". See Figure 1.
10. In the top field, enter the name of the device: “DMON_Host”. See Figure 3.
11. In the first two fields of USB ID, enter 0C55 and 1235. Leave the last USB ID field blank.
12. Above the “Install Driver” button, ensure that WinUSB is selected “WinUSB (v6.1.7600.16385)”
13. Click the “Install Driver” button.

14. Exit the program. You can now attach the DMON2 via the USB cable and launch d3host.

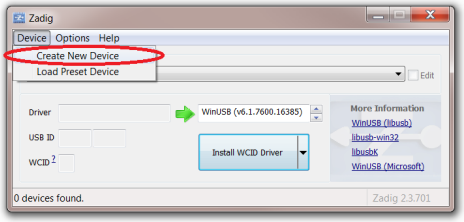


Figure 1. Create new USB device in Zadig

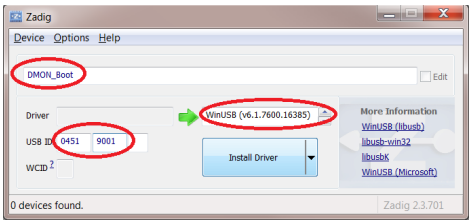


Figure 2. Create DMON_Boot device

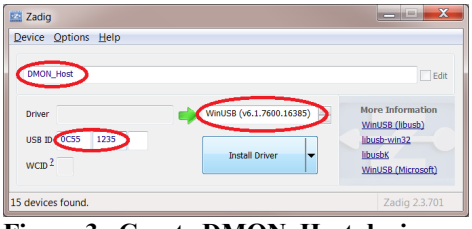


Figure 3. Create DMON_Host device

B. Connect to the DMON2 from the PC and download new firmware

1. Make sure the glider is shut down.
2. On the PC, make sure the caps lock is off (d3host is case sensitive) and turn up the speaker volume so you can easily hear the USB connect sound.
3. Connect the USB cable to the 8-pin bulkhead connector in the stiffener ring of the glider.
4. Plug the other end of the USB cable into the PC.
5. You should hear the USB-connect sound (two tones in rapid succession or a 4-note tune, depending on which version of Windows you are using) shortly after the USB cable is plugged into the PC.
6. If you don't hear the USB-connect sound, try the following:
    a. Plug the cable into a different USB port on the PC. Try each USB port in succession. Be sure to wait 10-15 seconds before unplugging, as the computer may be slow to recognize a new device on the USB port. Once you find a USB port that works, always connect to the DMON2 using that port.
7. Launch d3host.exe on the PC.
8. If d3host cannot find the DMON2 right away, it will display the message "DMON2 not found: try again?" Wait until you hear the USB connect sound a second time, then press "y" and d3host should connect to the DMON2. Upon connection, a message reading "Connected to DMON2 id xxxxxxxx" will be displayed, along with a menu of actions.
9. Select option 7 – "Download firmware" to put the custom firmware program in EEPROM.
    a. When prompted for the filename, type in the full pathname (e.g., C:\Baumgartner\NJ0420\LF2_HF240d1_15min.bin). Alternatively, if you copied the binary firmware file to the same directory where you launch d3host, you can just enter the filename (e.g., LF2_HF240d1_15min.bin).
    b. You should see "Erasing code space 100%", then "writing to FLASH……" several times, "Checking firmware in flash……", and finally "Successful firmware download!" if all goes well. You will see "Unable to open file XXXX" if the file cannot be found.
    c. Downloading the firmware need only occur once. The firmware is held in non-volatile memory, so it can be accessed every time the DMON2 starts up.
10. Select option 8 – "Current firmware" to verify the correct firmware was downloaded. Check that the correct binary file name is displayed, and the file/download times are appropriate.
11. Select option "q" – "Quit" to quit d3host.
    a. When prompted to "Run immediately?", respond "n" to shut down the DMON2.

C. Test the DMON2/LFDCS firmware and hydrophone prior to deployment.

1. Make sure the glider is shut down.
2. Use the instructions in section B above to connect the PC to the DMON2 via the USB cable and the d3host program.
3. Select option 3 – "Erase files" in d3host to clear the flash memory prior to testing.
    a. When asked "Are you sure you want to erase recordings?", respond "y".
    b. Select "a" to erase all files.
4. Select option "q" – "Quit" to quit d3host
    a. Upon quitting, select "y" when asked to "Run immediately?" This will instruct the DMON2 to begin running the firmware in EEPROM immediately after the USB cable is disconnected from the PC.
5. Disconnect the USB cable from the PC. The firmware will now begin to run on the DMON2. The red LED will flash inside the hydrophone head every 5 seconds for 30 minutes after the DMON2 is started. After 30 minutes, the LEDs will stop flashing, but the program will continue to run. The LEDs are intended to provide an external indicator that the firmware is running after startup.
6. Over the next few minutes, provide audio to the hydrophone that you will later recognize in the recordings (e.g., talk, play music, clap your hands, tap on the hydrophone, etc.). At the very least, tap on the hydrophone in some recognizable pattern (e.g., heartbeat: thump-thump-pause-pause-thump-thump-pause-pause…), as this signal will be recognizable later regardless of the background noise present.
7. After about a minute or so, reconnect the USB cable to the PC. Wait 30 seconds, then unplug the USB cable from the PC. Connecting the USB cable while the firmware is running will cause the program to gracefully shut down, a process that takes about 20 seconds. You should *not* hear the USB-connect sound when you connect the USB cable to the PC this first time. No USB connection will be made.
8. Plug the USB cable into the PC again. This time you should hear the USB-connect sound.
9. Launch d3host.
10. If you wish, select option "5" to change the directory into which offloaded files will be written. If you do not select this option, the files will be written into the same directory that contains d3host.exe. Once you select a new directory, it will be saved and all future offloads will be written to this directory.
11. Select 2 – "Offload files"
    a. Select "a" to offload all files (there should only be one file for this test)
    b. When prompted to "Enter filename base", type "XX". The DMON2 will proceed to copy the data in flash memory to the file XX001.dtg. If there are multiple files, it will offload the first to XX001.dtg, the second to XX002.dtg, the third to XX003.dtg, and so on.
12. Select "q" – "Quit".
    a. When prompted to "Run immediately?", respond "n" to shut down the DMON2.
13. Launch the PC program d3read.exe.
14. The program will display the directory where it thinks you are offloading DTG files. Verify that this directory is where you just put the offloaded XX001.dtg file. If it is not correct, type "y" when prompted to change. If the directory is correct, type "n".
15. When prompted for the "Input filename base", type "XX"

16. From the list of "Found files", select the file(s) that were previously offloaded (in this case, select XX001.dtg), or select "a" for all files if you want to unpack and decompress multiple files.
    a. Several files will be unpacked from any one DTG file. For the file XX001.dtg that you just created, d3read should unpack XX001.log, XX001.err, XX001.wav2, and XX001.xml (there may be XX001.wav1 and/or XX001.wav3 files as well, depending on the firmware you are using). The wav files have the recorded audio, the log file has messages stored by the DMON2 in an encoded format, the err file reports runtime errors, and the xml file has metadata.
17. Open XX001.log with a text editor and you should see something very much like the following:

```
1663867326,244536,$INIT,RTCRETRIES,0,0,0
1663867326,246063,$INIT,AUDIO,1,INIT_OK,AUDIO_ON,CHAN_ON,VTH_ON,VTH_OK,0021
1663867326,247580,$INIT,AUDIO,2,INIT_OK,AUDIO_ON,CHAN_ON,VTH_ON,VTH_OK,0021
1663867326,249101,$INIT,AUDIO,3,INIT_OK,AUDIO_ON,CHAN_ON,VTH_ON,VTH_OK,0021
1663867326,258587,$INIT,HEADINIT,INIT_OK,38,38,0
1663867326,258681,$INIT,HEADRETRIES,0
1663867326,265070,$INIT,CHANINIT,2,OK,2000
1663867326,270740,$INIT,CHANINIT,4,OK,240000
1663867326,270824,$INIT,ALLJOBINIT,OK
1663867326,270997,$INIT,IICRETRIES,159,0,0
1663867328,7154,$START,632C99C000000E231E2C00010000*1B
1663867362,96,$DUTY,632C99C00004632C99DF00052000000010000F618*0A
1663867364,6209,$END,632C99E400060E231E2C1*20
1663867366,38,LOGEND,0000000000……0000000000
1663867366,274,LOGEND, 0000000000……0000000000
1663867366,390,LOGEND, 0000000000……0000000000
1663867366,1072,LOGEND, 0000000000……0000000000
1663867366,1188,LOGEND, 0000000000……0000000000
```

    a. The first two fields indicate the time, so these will be different for your test.
    b. The important thing is that the XX001.log file has $INIT strings, a $START string, and an $END string. There will be a number of LOGEND strings at the end of the file with lots of zeroes at the end of each string. The number of LOGEND strings will likely change if you do this test again and again. This is not a problem.
18. Open each of the wav files with a media player and play the audio. You should recognize the sounds you made/played to the glider hydrophone (note that they will sound muffled because of the hydrophone's urethane housing, which is acoustically transparent in water but not in air). If you examine the sound using software that allows you to plot spectrograms (e.g., Raven), do not be alarmed if there is a tremendous amount of noise in the recording. Ships and some labs are notoriously noisy (both acoustically and electrically), so there may be lots of noise present in the recording. This test is simply to make sure signals from the hydrophone are being recorded and processed by the DMON2/LFDCS firmware. Any local noise will, of course, disappear once the glider is in the water. The multi-channel recorder firmware will produce wav files with the following extensions: .wav1 for recordings from the mid-frequency hydrophone (0.1 – 50 kHz

bandwidth, -200 dB re V/µPa rms sensitivity), .wav2 for recordings from the low-frequency hydrophone (10 – 7500 Hz bandwidth, -203 dB re V/µPa rms sensitivity), and .wav3 for recordings from the high-frequency hydrophone (1 – 160 kHz bandwidth, -206 dB re V/µPa rms sensitivity). Note that every time the red LED flashes (for the first 30 minutes after startup), there will be a corresponding impulsive noise in each recording caused by electrical noise. This is normal and one of the reasons the red LED only flashes for the first 30 minutes.

D. Charge the DMON2 battery

1. Make sure the glider is shut down.
2. Use the instructions in section B above to connect the PC to the DMON2 via the USB cable and the d3host program.
3. Once connected in d3host, select 4 – "Battery".
4. Make sure "Battery connected" is reported and note the battery voltage displayed just to the right of "Battery connected". If the battery voltage is less than 4 volts, keep the USB cable attached to the hydrophone, which will allow the DMON2 battery to charge via the USB port on the PC. Depending on how depleted the battery is, it may take several hours to fully charge the battery. If possible, set the computer up to charge the DMON2 battery overnight. Make sure your computer does not go to sleep during the charging process, as this will almost surely turn off the power to the USB port. Alternatively, you can attach the USB cable to a USB charger that plugs into an electrical outlet (e.g., the same chargers that are used with smartphones) and leave it plugged in overnight.
5. Check the battery voltage at any time in d3host using option 4 – "Battery".
6. When fully charged, the battery will have a voltage of 4.1 V or more. A depleted battery will have less than 3.8 V. It is good practice to fully charge the DMON2 battery prior to deployment. This is not necessary, but the glider's battery capacity will be reduced (i.e., wasted) by charging up the DMON2 battery when it is first deployed. It is recommended that the DMON2 battery have a voltage of 4 V or more prior to deployment.
7. When done charging, select option "q" – "Quit" in d3host and when prompted to "Run immediately?", respond "n".

E. Prepare the DMON2 for deployment

1. Make sure the glider is shut down.
2. On the PC, set the computer time zone to GMT, and then set the computer time to the exact GMT time. Before exiting, you will set the DMON clocks using the PC clock, so you will want to ensure that the PC clock is accurately set to GMT time.
3. Use the instructions in section B above to connect the PC to the DMON2 via the USB cable and the d3host program.
4. Select option 3 – "Erase files" in d3host to clear the flash memory prior to deployment.
    a. When asked "Are you sure you want to erase recordings?", respond "y".
    b. Select "a" to erase all files. The DMON2 should have no files in flash memory prior to deployment.
5. Select option 4 – "Battery" to verify that the battery is at least 4 V.
6. Select option 8 – "Current firmware" to verify the correct firmware will be used for the deployment. Check that the correct binary file name is displayed, and the file/download times are appropriate.
7. Select option 9 – "Set real-time clocks" to set the onboard and Dallas real-time clocks to match the PC's clock.
8. Select option "q" – "Quit" to quit d3host.
    a. When prompted to "Run immediately?", respond "n". This will shut down the DMON2 and it will remain shut down until external power is applied to the DMON2 board. When the glider is started, it will apply power to the board and the DMON2 will automatically run its firmware without any user intervention.
9. Disconnect the USB cable from the external glider hydrophone and attach the dummy plug. The DMON2 is now ready for deployment.

F. Deployment

1. No special action is required when the glider is powered up. Once the glider supplies power to the DMON2, it will wake up and begin running the firmware in non-volatile memory. Upon startup, the LEDs inside the hydrophone head will begin to flash once every 5 seconds. They will continue to flash for the next 30 minutes, after which they will stop flashing; even though the LEDs are no longer flashing, the firmware is still running. The LEDs are intended to provide an external indicator that the firmware is running after startup.
2. *Optional:* Before putting the glider in the water, produce a series of impulsive noises near the glider at known times that can later be used to verify the drift of the DMON2 clock. Try to do this in as quiet an environment as possible (e.g., not while the ship is steaming).
    a. Find something that can be used to make an impulsive noise. A wrench wacked against a pipe or the ship's deck is perfect.
    b. Using a GPS that is reporting GMT time, make 3 impulsive sounds near the glider spaced 1 second apart. Write down the exact time of the first of the 3 sounds as reported by the GPS.
    c. Some time later (say, 15-30 seconds later), make 2 impulsive sounds near the glider spaced 1 second apart. Write down the exact time of the first of the 2 sounds as reported by the GPS.
    d. Some time later (say, 15-30 seconds later), make 1 impulsive sound near the glider. Write down the exact time of the sound as reported by the GPS.
3. From the perspective of the DMON2, the glider is now ready to deploy.
4. Once deployed and on mission, the glider will store DMON2 serial strings sent from the DMON2 to the glider in a segment file ending in .asc. These data can be offloaded from the glider during each Iridium communication session if desired to verify that the DMON2 is working, but this is not required.

G. Recovery

1. *Optional:* Shortly after recovery (while the DMON2 is still running), produce a series of impulsive noises near the glider at known times that can later be used to verify the drift of the DMON2 clock. Try to do this in as quiet an environment as possible (e.g., not while the ship is steaming).
    a. Find something that can be used to make an impulsive noise. A wrench wacked against a pipe or the ship's deck is perfect.
    b. Using a GPS that is reporting GMT time, make 3 impulsive sounds near the glider spaced 1 second apart. Write down the exact time of the first of the 3 sounds as reported by the GPS.
    c. Some time later (say, 15-30 seconds later), make 2 impulsive sounds near the glider spaced 1 second apart. Write down the exact time of the first of the 2 sounds as reported by the GPS.
    d. Some time later (say, 15-30 seconds later), make 1 impulsive sound near the glider. Write down the exact time of the sound as reported by the GPS.
2. Once the glider and hydrophone are hosed down, bring the glider to a sheltered area where you can safely attach the USB cable to the hydrophone and have a PC nearby.
3. Although the DMON2 will start when the glider is started, the DMON2 will not shut down when the glider is shut down. You must send it a signal to shut down. This is simply done by connecting the glider hydrophone to the PC via the USB cable. Wait 30 seconds, then unplug the USB cable from the PC. Connecting the USB cable while the firmware is running will cause the program to gracefully shut down, a process that takes about 20 seconds. You should *not* hear the USB-connect sound when you connect the USB cable to the PC this first time. No USB connection will be made.
4. Plug the USB cable into the PC again. This time you should hear the USB-connect sound.
5. Launch the d3host program.
6. If you wish, select option "5" to change the directory into which offloaded files will be written. If you do not select this option, the files will be written into the same directory that contains d3host.exe. Once you select a new directory, it will be saved and all future offloads will be written to this directory.
7. Select 2 – "Offload files"
    a. Select "a" to offload all files
    b. When prompted to "Enter filename base", enter the name of your mission. The DMON2 will use this as the file prefix (e.g., if you enter "nj0420_", then the DTG files will be saved as "nj0420_001.dtg", "nj0420_002.dtg", "nj0420_003.dtg", etc.).
    c. The DMON2 will likely take a long time (several hours) to offload all the files from a several week- to several month-long deployment.
8. Select "q" – "Quit".
    a. When prompted to "Run immediately?", respond "n" to shut down the DMON2.
    b. *It is recommended that you do not erase the files from the DMON2 until all the data are backed up elsewhere. Even then, leave the data on the DMON2 until they absolutely have to be erased.*
9. Launch the PC program d3read.exe.

10. When prompted for the “Input filename base”, type the same filename base you used when offloading the data in d3host (e.g., “nj0420_” in the example above).
11. Select “a” for all files to unpack and decompress multiple files.
    a. Unpacking and decompressing will also take a long time (several hours.
12. Backup all files (*.wav, *.log, *.xml) to an external hard drive as soon as possible.

H. Installing new _boot.bin and _reboot.bin

There is a very small chance that the PC you have does not have the right files to correctly communicate with the DMON2, or to launch new firmware. If so, you may have to use the following instructions.

1. Copy _boot.bin and _reboot.bin files to the same folder with d3host.exe and d3read.exe.
2. Use the instructions in section B above to connect the PC to the DMON2 via the USB cable and the d3host program.
3. Select option 6 – "Enter expert mode"
4. Select option 6 – "EEPROM write"
5. Enter _reboot.bin when prompted for the filename
6. Enter 0 when prompted for the address
7. Select option "q" – "Quit" to quit.
8. Disconnect USB cable from PC to force the DMON2 to power down.

I. Format of DMON2 messages

Messages sent from the DMON to the glider and also saved in flash memory in the *.log file are formatted in NMEA-like strings, with a "$" at the beginning, followed by a message name (e.g., START), followed by a data payload ending in "*", and finally a checksum. The payload consists of human-readable hexadecimal numbers comprising several data fields that are concatenated. Example messages are shown and decoded below.

Duty-cycled recordings are saved in concatenated wav files. For example, consider a recording with a 1 minute every 15 minutes duty cycle. The first 1-minute recording (recording session #1) will begin 15 minutes after startup and end 1 minute later. The second 1-minute recording (recording session #2) will begin 30 minutes after startup and end 1 minute later. In the wav file, these two 1-minute files will be concatenated (i.e., the first sample of session #2 will immediately follow the last sample of session #1). The $DUTY messages in the *.log file are used to separate the audio associated with each recording session. Note that for multi-channel recordings, there will be $DUTY messages for each hydrophone. $DUTY messages are also generated for continuously recorded audio, but there is no need to separate out recording sessions, since they are appropriately contiguous in the wav files.

*$START,632C99C000000E231E2C00010000*1B*

- 632C99C0 – date/time (seconds since January 1, 1970 00:00:00)
- 0000 – milliseconds after whole second value
- 0E231E2C – DMON2 board id
- 00010000 – flash memory size in MB
- 1B – checksum

*$DUTY,632C99C00004632C99DF00052000000010000F618*0A*

- 632C99C0 – start date/time of recording session (seconds since January 1, 1970 00:00:00)
- 0004 – milliseconds after whole second value
- 632C99DF – end date/time of recording session (seconds since January 1, 1970 00:00:00)
- 0005 – milliseconds after whole second value
- 2 – hydrophone (1 – mid-frequency, 2 – low-frequency, 3 – high-frequency)
- 00000001 – sequential recording session number
- 0000F618 – number of samples recorded

*$END,632C99E400060E231E2C1*20*

- 632C99E4 – date/time (seconds since January 1, 1970 00:00:00)
- 0006 – milliseconds after whole second value
- 0E231E2C – DMON2 board id
- 1 – shutdown code
- 20 – checksum

J. DMON2 lithium-ion batteries

The DMON2 has an integrated lithium-ion rechargeable battery that serves as a backup to an external power source. When the DMON2 is not running its firmware and is in hibernation mode, it will also draw a very small amount of power from the lithium battery to keep some basic wake-up and clock functions operational. On rare occasions, the battery voltage may drop too low for the DMON2 to recharge it, and d3host will report that there is no battery connected to the DMON2. In such cases, the battery may be able to be revived with a lithium-ion battery charger, such as a Tenergy TB6B. You can test the lithium battery at its connector by measuring the voltage across the red and black wires that are positioned opposite one another in the connector body. Be VERY CAREFUL when doing this to NOT short the battery (i.e., connect the red and black wires). Based on the voltage, do the following:

~ 0.0 V – battery protection board is not passing voltage (see below)
< 2.5 V – discard
2.5-3.6 V – revive with Tenergy TB6B battery charger
> 3.6 V – charge with DMON2

If the battery will be connected to the DMON2 during storage, fully charge the battery prior to storage. This is routinely done while downloading data from the last mission. If storing the battery independent of the DMON2 (i.e., not connected to the DMON2), there is no need to fully charge the battery.

The lithium battery has an attached battery protection board (beneath the tape at the end of the battery from which the wires emerge) that does not allow any voltage to pass if the battery is below 2.5 V. If you test the voltage on a battery and get 0 V, it is very likely that the battery protection board is not passing voltage. Such a battery may still be tested and possibly revived if you remove the protective tape and expose the gold and silver tabs that attach to the battery protection board. Test the voltage across these tabs, and if it is not too low, try to recharge the battery using a Tenergy TB6B battery charger (or equivalent).

K. Shutting down a DMON2 instrument without attaching a USB cable

There are times when it is helpful to shut down a glider-installed DMON2 remotely. The following instructions will put the DMON2 into its very low-power hibernation mode during which it will not record or process audio, nor communicate with the glider. This process is equivalent to attaching a USB cable to the DMON2 when the firmware is running to gracefully shut the DMON2 down, or by choosing "n" to the question "Run immediately?" when quitting d3host.

1. In GliderDOS, type "consci" to use the science computer
2. Use the command "u4stalk" to connect directly to the DMON
3. Once connected to the DMON2, type the "$ADSTOP" and hit the return key (do not include the quotes)
4. The DMON should respond within a few seconds with a message that starts with "$END", which indicates it has started its graceful shutdown (which takes on the order of 10-15 seconds)
5. Type ctrl-C to exit u4stalk
6. Type "quit" to return to GliderDOS
7. If you are shutting down the DMON2 to preserve as much glider battery as possible, you can now type "use - science_super" to shut down the science bay computer. If you are shutting down the glider, you can now type "exit". If you do not do this, then the next time the science bay powers the DMON2, it will start up and run its firmware, and you will need to either attach a USB cable or type $ADSTOP in u4stalk again to shut the DMON2 down again.

L. DMON2 hydrophone information

The following are the factory sensitivities and bandwidths of the 3 DMON2 hydrophones:

Low-frequency: -203 dB re: V/µPa rms, 10 Hz – 7.5 kHz
Mid-frequency: -200 dB re: V/µPa rms, 100 Hz – 50 kHz
High-frequency: -206 dB re: V/µPa rms, 1 kHz – 160 kHz

For each hydrophone, there is a gain of 33.2 dB (20 dB preamp gain, 13.2 dB main board gain), the zero-to-peak V_{adc} is 1.5 V, and the system is 16 bits. That should be all you need to calculate sound pressure level and/or power spectral density from the digital recordings using the equations in the following publication (and supplemental material):

Merchant, N.D., Fristrup, K.M., Johnson, M.P., Tyack, P.L., Witt, M.J., Blondel, P. and Parks, S.E. (2015), Measuring acoustic habitats. Methods in Ecology and Evolution, 6: 257-265. https://doi.org/10.1111/2041-210X.12330.